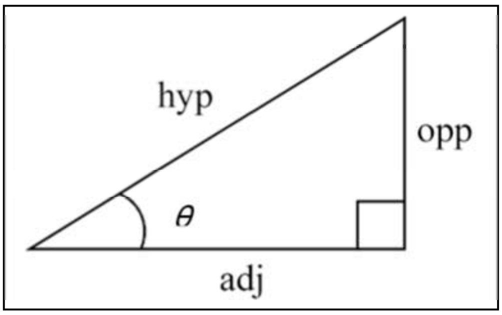
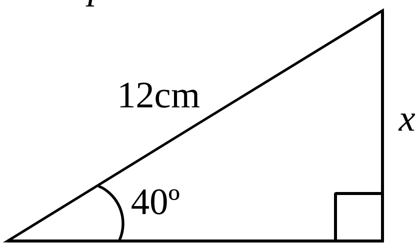
Worksheet A6: Calculating with Sine



There are 3 different calculations we can make. We can find the missing opposite side, hypotenuse or angle, if we know the other two.

$$\sin\theta = \frac{\text{opp}}{\text{hyp}}$$

Example 1



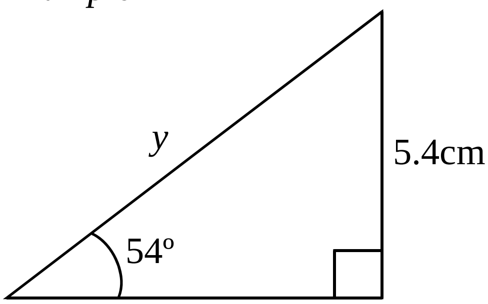
$$\sin\theta = \frac{\text{opp}}{\text{hyp}}$$

$$\sin 40 = \frac{x}{12}$$

$$12 \times \sin 40 = x$$

$$x = 7.7\text{cm} \quad (\text{to 1d.p.})$$

Example 2



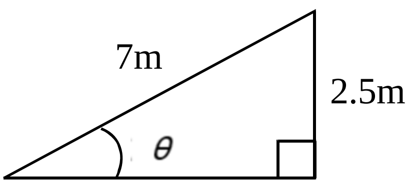
$$\sin\theta = \frac{\text{opp}}{\text{hyp}}$$

$$\sin 54 = \frac{5.4}{y}$$

$$y = \frac{5.4}{\sin 54}$$

$$x = 6.7\text{cm} \quad (\text{to 1d.p.})$$

Example 3



$$\sin\theta = \frac{\text{opp}}{\text{hyp}}$$

$$\sin\theta = \frac{2.5}{7}$$

$$\sin\theta = 0.357$$

$$\theta = 20.9^\circ \quad (\text{to 1d.p.})$$

Use the $\sin^{-1}$ button on your calculator

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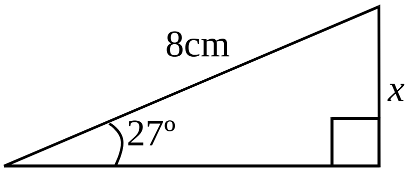
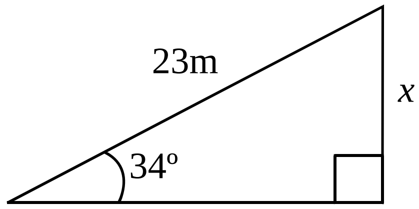
Exercise

1. Find the size of the side *x* to 1 decimal place.

(a)

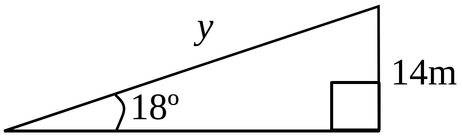


(b)

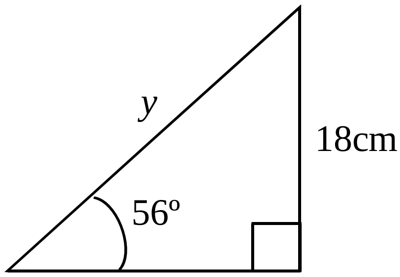


2 . Find the size of the side *x* to 1 decimal place.

(a)

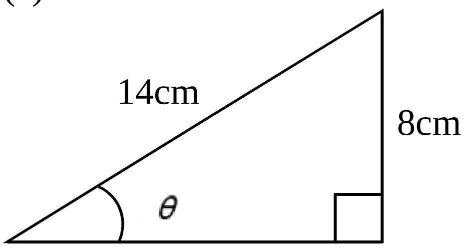


(b)

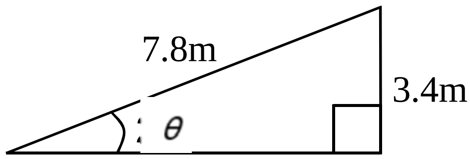


3. Find the size of the angle θ to 1 decimal place.

(a)

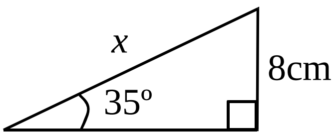


(b)

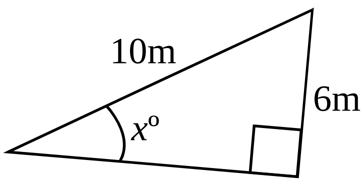


4. Find *x* in each case to 1 decimal place.

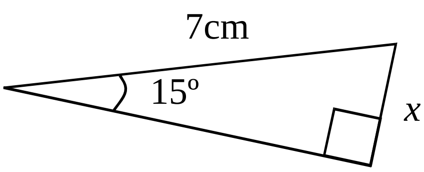
(a)



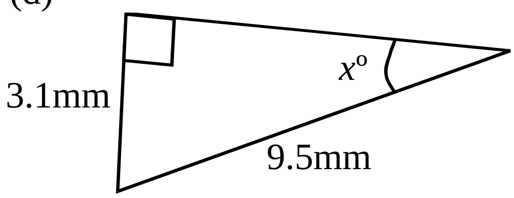
(b)



(c)



(d)



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